

Use algebra to model relationships efficiently

Name: _____

Date: _____

Step 11

Middle School



Today I am learning to use algebra to model real-world relationships and make predictions.

1. Write the Rule

Look at each table. Write the rule.

Input	1	2	3	4	5
Output	4	5	6	7	8

 Rule: Output = $n +$ _____

Input	1	2	3	4
Output	2	4	6	8

Rule: Output = _____

2. Use the Rule

Use the rule to complete each table.

 Rule: Output = $n + 3$

n	1	2	3	4	5
Output	_____	_____	_____	_____	_____

 Rule: Output = $2n$

n	1	2	3	4	5
Output	_____	_____	_____	_____	_____

3. Match the Situation

Draw a line to match each situation with the correct rule.

Situations

- 1 A taxi charges \$5 plus \$2 per kilometre.
- 2 A movie ticket costs \$12 each.
- 3 A plant grows 3 cm each week.
- 4 A savings account starts at \$10 and gains \$5 each week.

Rules

- Cost = $12n$
- Height = $3n$
- Cost = $5 + 2n$
- Savings = $10 + 5n$

4. Build the Model

Write a rule and complete each table.

- a) Each packet contains 4 stickers.

Rule: Stickers = _____

Packets	1	2	3	4
Stickers	_____	_____	_____	_____

- b) A game costs \$8 to enter plus \$2 per ride.

Rule: _____

Rides	0	1	2	3
Cost (\$)	_____	_____	_____	_____



5. Find the Missing Value

Use the rule to find y.

a) $y = n + 4$

If $n = 7$

$y =$ _____

b) $y = 3n$

If $n = 5$

$y =$ _____

c) $y = 2n + 1$

If $n = 4$

$y =$ _____

d) $y = n + 10$

If $n = 12$

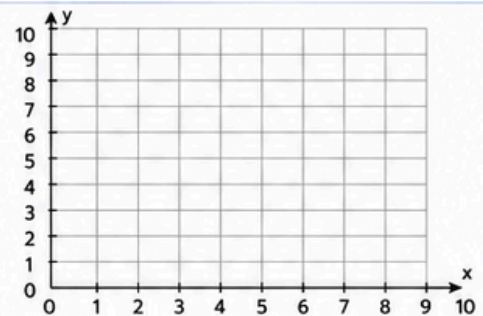
$y =$ _____

6. Graph the Relationship

Use the rule to complete the table and plot the points on the graph.

 Rule: $y = n + 2$

n	1	2	3	4
y	_____	_____	_____	_____



7. Real-Life Algebra

A fundraiser sells cupcakes. Each cupcake costs \$3.

 Rule: Money = $3n$


- a) How much money is raised if 5 cupcakes are sold?
Answer: \$ _____
- b) How much money is raised if 10 cupcakes are sold?
Answer: \$ _____
- c) How much money is raised if 20 cupcakes are sold?
Answer: \$ _____

8. Think and Explain

Write your answers in complete sentences.



- a) Why is algebra useful?



- b) How does a rule help us predict values?



- c) How is a table connected to an algebra rule?



I can...

- ☐ use algebraic rules to represent relationships.
- ☐ make tables and graphs from rules.
- ☐ solve real-life problems using algebraic models.



My Learning Goal

I can use algebra to model, solve, and understand relationships efficiently.

